

Subject Description Form

Subject Code	FSN2416
Subject Title	Introduction to Food Science
Credit Value	3
Level	2
Pre-requisite	Nil
Exclusion	FSN1D08 and FSN1D10
Objectives	The subject is intended to introduce the basic elements of food science, including the chemical composition of food, food microbiology and safety as well as food preservation and production technology.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. define the major nutrients and chemical components in food; b. describe the key foodborne hazards and the control measures; c. relate the use of technology in food production; d. develop analytical, critical thinking, and written communication skills.
Subject Synopsis/ Indicative Syllabus	Basics of Food Science: Overview of food science; food supply chain; raw food materials; food chemistry on macro and micro-nutrients; food microbiology and food processing methods. Basics of Food Safety: Introduction on Food chemical and biological hazards; Use of food additives; Food Quality assurance and control. Basics of Food innovation: Food biotechnology; product development and sensory science.
Teaching/Learning Methodology	The basic contents of this subject will be presented with the aid of lecture notes, online videos, LMS platform and other teaching tools. Small-group discussions will be held in class to supplement lecture for students to relate food science knowledge in daily practice. Students are also expected to study reference materials distributed in class, from the library or other sources (e.g. newspaper and magazine clippings, and information available on the Internet). A variety of assessment tools will be used, including quizzes and assignments to develop students' analytical skills, critical thinking and communication skills. Laboratory sessions on basic food science are designed to facilitate students' learning.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1. Lab Reports	10%	✓		✓	✓
	2. In-class activities/ Assignment	30%	✓	✓	✓	✓
	3. Test	20%	✓	✓	✓	✓
	4. Examination	40%	✓	✓	✓	✓
	Total	100 %				
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Lab reports provide students with the opportunity to apply theoretical knowledge in practical settings, fostering critical thinking and analytical skills as they document and interpret experimental results. In-class assignments and activities encourage active participation and collaborative learning, reinforcing key concepts and promoting problem-solving abilities. The test serves as a formative assessment to gauge students' grasp of foundational principles and their ability to apply them in various contexts. Finally, the comprehensive final examination evaluates the students' overall mastery of the subject matter, integrating knowledge from across the course to assess their readiness for advanced study or professional practice in the field of food science.						
Student Study Effort Required	Class contact:					
	▪ Lecture/ Seminar				33 Hrs.	
	▪ Lab				9 Hrs.	
	▪ Other student study effort:					
	Self study				82 Hrs.	
	▪ Total student study effort				124Hrs	
Reading List and References	1. Shewfelt, Robert L. Introducing food Science. Boca Raton: CRC press, 2009					
	2. Vaclavik, Vickie A. & Christian, Elizabeth W. Essentials of Food Science, 3 rd edition. Springer, 2008					
	3. McWilliams, Margaret. Food Fundamentals, 10 th edition. Pearson, 2013					